Screening Financial Disclosure Risk through Textual-Financial Feature Fusion and Explainable Anomaly Detection: Evidence from Chinese A-share Manufacturing Firms, 2021–2023

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Abstract

This proposal develops an explainable anomaly-screening framework for annual reports issued by Chinese A-share manufacturing firms from 2021 to 2023. It compares finance-only, text-only, and textual-financial feature matrices under a common sample and evaluation design. Six financial indicators capture profitability, leverage, growth, operating cash flow, receivables efficiency, and accrual-related risk. Three MD&A indicators capture readability, sentiment balance, and uncertainty. Isolation Forest is the primary unsupervised model, and an autoencoder provides a robustness ranking. SHAP-based attribution will explain score drivers globally and for selected firm-year observations. Rankings will be evaluated against the Beneish M-Score and, subject to transparent time matching, an external outcome benchmark such as an inquiry letter, enforcement action, restatement, or modified audit opinion. Scores are review-priority signals, not direct measures of disclosure quality, fraud, misconduct, or violations. The study contributes a transparent and reproducible test of the incremental screening value of Chinese MD&A information beyond structured financial data. On the final sample of 8,140 firm-years, the fused ranking correlates weakly but significantly with the Beneish M-Score (Spearman rho 0.103, $p < 0.001$); the finance-only ranking correlates more strongly (0.136), and adding the three MD&A textual features does not improve the ranking, while the top-10% review list enriches Beneish flags by a factor of 1.18.

Keywords: Financial disclosure quality; Chinese MD&A; textual-financial feature fusion; anomaly screening; Isolation Forest; explainable artificial intelligence

1. Introduction

Corporate disclosure reduces information asymmetry between firms and external stakeholders, making it essential for decision-making in capital markets[1]. However, disclosure quality is multi-dimensional and not directly observable; attributes such as earnings characteristics, readability, restatements, audit outcomes, and regulatory actions reflect different aspects of reporting and are not interchangeable[2].

Annual financial reports contain both structured financial information and management narratives. As the length and complexity of these reports increase, comprehensive manual review becomes progressively more challenging [3]. The Management Discussion and Analysis (MD&A) section is particularly informative because it contextualizes firm performance, risks, and prospects alongside the

financial statements. Previous studies link annual-report readability to earnings characteristics and information-processing costs [4–6]. However, due to differences in language and structure, readability metrics developed for English cannot be directly applied to Chinese annual reports [7]. Therefore, this study employs readability, tone, and uncertainty indicators suitable for Chinese, and combines them with financial metrics covering profitability, leverage, growth, cash flow, operational efficiency, and accrual-related risks, to assess whether MD&A information can provide incremental screening value beyond structured financial data.

A further challenge is the absence of complete and consistent disclosure-quality labels. Regulatory enforcement actions, restatements, and modified audit opinions provide useful external signals but capture different outcomes and remain incomplete. This motivates an unsupervised screening approach. Isolation Forest is used to rank unusual firm-year observations without requiring complete labels [8,9], while an autoencoder provides a complementary robustness check based on reconstruction error [10]. The Beneish M-Score is used as an accounting-risk benchmark rather than as a direct measure of disclosure quality [11,12].

Explainability is essential if anomaly rankings are to support review prioritization. SHAP is therefore used to examine how financial and textual variables contribute to anomaly scores at both the global and firm-year levels [13].

Against this background, this study investigates three questions: whether the features of Chinese Management Discussion and Analysis (MD&A) can provide screening value beyond financial variables; which groups of financial and textual features contribute most to the anomaly scores; and whether the resulting rankings are robust across sample years, manufacturing subindustries, and alternative model specifications. The contribution of this study lies in integrating Chinese narrative indicators with structured financial information, conducting text evaluation through comparable finance-only, text-only, and combined models, and combining unsupervised screening with transparent feature attribution.The proposed framework is designed for review prioritization. The anomaly scores serve to identify observations worthy of further scrutiny, rather than acting as an absolute measure of disclosure quality or evidence of misconduct.

2. Literature Review

Previous studies emphasize that financial disclosure quality is multidimensional and cannot be adequately captured by a single indicator [14]. Textual attributes such as readability and tone provide additional information [15], but due to language differences, quantitative indicators in English cannot be directly applied to Chinese MD&A. Therefore, Chinese financial statement analysis requires text-based indicators and resources that are suitable for the language.

Structured financial variables and MD&A disclosures capture complementary aspects of corporate reporting [16]. Accounting variables summarize realized financial outcomes, whereas MD&A

narratives provide contextual information on performance, risks, and future prospects. Consistent with the multidimensional perspective in the earnings-quality literature, narrative information may provide complementary information beyond structured accounting data. However, it is only worth including if it provides greater screening value than financial variables, and its incremental value should be comprehensively assessed by comparing models that use only financial data, only textual data, and a hybrid approach.

In the absence of complete disclosure-quality labels, unsupervised anomaly detection provides a practical screening approach. Isolation Forest [17] identifies unusual observations through random partitioning, while Autoencoders [18] detect deviations based on reconstruction error. Their different mechanisms allow the robustness of anomaly rankings to be examined across alternative model specifications.Because anomaly scores measure statistical abnormality rather than misconduct, their economic relevance should be assessed against external benchmarks. The Beneish M-Score is used as an accounting-risk benchmark, while regulatory enforcement actions and financial restatements serve as external outcome indicators [19,20]. SHAP-based feature attribution is further used to improve explainability by identifying whether anomaly scores are primarily driven by financial variables, textual features, or both.This facilitates bridging the gap between statistical anomaly detection and actionable regulatory decisions, but it does not directly suggest causal relationships or provide explanations for misconduct[21].

In summary, three gaps remain. First, evidence on whether Chinese MD&A provides incremental screening value beyond structured financial variables remains limited. Second, distinctions among statistical anomalies, accounting-risk proxies, and observed enforcement outcomes are not always explicit. Third, an integrated and explainable screening framework for the Chinese A-share reporting environment remains underdeveloped.To address these gaps, this study develops an explainable anomaly-screening framework that integrates structured financial variables with Chinese MD&A features. Incremental textual value is evaluated through financial-only, text-only, and fused models, while robustness and explainability are examined using alternative anomaly detectors, SHAP attribution, and external benchmarks. The framework is intended for review prioritization rather than misconduct determination.

3. Research Design and Methodology

3.1 Research Scope, Sample, and Data Sources

The unit of analysis is the firm-year. The sample covers Chinese A-share firms classified as manufacturing for fiscal years 2021, 2022, and 2023. The sample is restricted to manufacturing firms to improve cross-sectional comparability while preserving meaningful firm-level variation. The industry-classification rule, the handling of classification changes, and the sample-selection flow will all be settled before estimation. A firm-year qualifies only when its annual report is accessible, the

MD&A section is identifiable, and the core financial inputs are complete. Rather than imputing missing values in silence, we will report how many observations are lost to unavailable reports, failed text extraction, failed company-year matching, or missing core values.

Financial variables come from a traceable database, either CSMAR or Wind, and annual reports together with MD&A text are collected from CNINFO or another authoritative disclosure source. For every observation we keep the stock code, fiscal year, report version, source location, and extraction date, so that each firm-year can be traced back to the documents behind it. External evaluation variables comprise a documented Beneish M-Score and, where data coverage and timing allow, inquiry letters, enforcement actions, financial restatements, and modified audit opinions. None of these variables enters model fitting, and none is treated as complete ground truth for disclosure quality.

3.2 Textual-Financial Feature Construction

The finance-only matrix uses six indicators chosen in advance: return on assets, the ratio of total liabilities to total assets, revenue growth, operating cash flow over total assets, an accounts-receivable-turnover measure, and total accruals over total assets. Before estimating any model, we will compile a feature dictionary that fixes the formulae, accounting dates, source fields, denominator treatment, outlier handling, and standardisation rules for each indicator.

The text-only matrix has three MD&A indicators. The first is a Chinese readability measure built on a documented length-and-complexity rule; the second is a sentiment-balance score derived from positive and negative financial language; the third is an uncertainty ratio based on a documented method for Chinese financial language. We will log the section-location rule, text cleaning and segmentation steps, model or dictionary version, thresholds, and the handling of failed extraction, so that the text processing can be reproduced.

Financial outliers are handled with a winsorisation rule set in advance, after which all numerical features are standardised within manufacturing subindustry-year groups wherever the sample is large enough, with a documented fallback for sparse cells. The finance-only, text-only, and fused nine-feature matrices are built from the same eligible firm-years and follow the same preprocessing logic. Holding the sample and preprocessing constant means that holding the sample and preprocessing constant isolates the effect of adding textual features on anomaly rankings; whether these changes represent incremental screening value is evaluated using external benchmarks.

3.3 Unsupervised Anomaly Screening

Isolation Forest serves as the primary model because it can rank statistically unusual observations without complete supervised labels. We run it with 500 trees, a maximum subsample of 256 observations (or the whole sample when smaller), a fixed random seed, and a top-10% review-priority threshold chosen in advance. The direction of the score is checked and recorded, so that a larger score

always indicates greater statistical abnormality. The parameters are not tuned against enforcement, restatement, or audit outcomes.

We estimate the model separately on the finance-only, text-only, and fused matrices. The output is a continuous anomaly score and rank rather than a definitive abnormal-or-compliant label. A high score only moves an observation further forward in the queue for human review; it does not by itself establish poor disclosure, fraud, or a regulatory violation.

3.4 Robustness Model and Explainability

An autoencoder offers a second ranking, based on reconstruction error, and is trained on the same three feature matrices. Before interpreting its results, we will record the architecture, bottleneck dimension, activation functions, regularisation, optimiser, number of epochs, early-stopping rule, batch size, and random seed. The point of this exercise is to see whether the main ranking patterns survive a different detection mechanism; the autoencoder is not used to assign a definitive strong-or-weak anomaly label.

We attribute the Isolation Forest scores with a validated SHAP-compatible method.The explained output is oriented so that higher values consistently indicate greater abnormality. At the global level, we report feature importance and the relative contribution of the financial and textual feature groups. At the local level, selected highly ranked firm-years are presented together with their raw and standardised values, textual measures, source provenance, and feature contributions. SHAP tells us how the chosen model arrives at a score; it neither identifies causal mechanisms nor explains why misconduct occurred.

4. System Implementation and/or Model Development

4.1 Pipeline Overview and Reproducibility

Section 3 describes what we plan to do; this section describes what has actually been built so far and how it works. The whole study is implemented as an ordered set of Python scripts rather than a single large program. Each script has one clear job: it reads one input file, performs one step, and writes one output file. This makes the system easy to check, easy to debug, and easy for another person to run again. The scripts are organised into folders for data ingestion, feature construction, models, and reporting, so the responsibilities are separated in an obvious way.

The pipeline runs through eight main stages in a fixed order. First, the sample is constructed from the CSMAR basic-information table. Second, the three financial statements are parsed into one intermediate table. Third, the downloaded annual-report files are checked against the sample list. Fourth, the MD&A text is extracted from each PDF. Fifth, a quality-control report flags missing or unusual records. Sixth, the finance and text features are combined into one feature table. Seventh, descriptive statistics summarise the sample. Eighth, the models are run and the results are narrated into a readable report. A

single script can run this whole chain, and each stage can also be run on its own when only part of the pipeline needs to be repeated.

Reproducibility is treated as a first-class requirement rather than an afterthought. Every random process uses the same fixed seed (42), so repeating a run on the same data gives the same numbers. All parameter choices are written in one configuration file, and all intermediate results are saved as plain CSV files that can be inspected directly. The code is covered by automated tests, which currently number more than seventy and all pass.

4.2 Data Acquisition and Quality Control

The sample is built from the CSMAR listed-company basic-information annual table. We keep only manufacturing firms, which we identify with the industry code that starts with the letter “C”, and we keep the study years 2021, 2022 and 2023. The year 2020 is also kept as a base year because some features, such as revenue growth and the Beneish M-Score, need the previous year's numbers. After filtering, the sample contains about 3,700 firms and roughly 10,000 firm-year observations.

The financial data are taken from three CSMAR consolidated statements: the balance sheet, the income statement, and the cash-flow statement. Only the fields needed by the six financial features are read, which keeps the memory use manageable even though the original files are several hundred megabytes. The annual-report texts are obtained from the CNINFO disclosure website. For each firm-year we download the annual report PDF for the correct disclosure year.

Downloading thousands of PDFs is where most practical problems appear, so we added a download audit step. After downloading, every file is checked against the sample list, and the outcome is recorded as either usable, missing, or mismatched. The audit for the current run recorded 10,051 usable firm-year files and 121 missing ones. Most of the missing cases are firms that were not yet listed, or had already been delisted, in that year, so they are a data-availability boundary rather than a software failure.

The text step extracts the management discussion and analysis (MD&A) section from each PDF using the pypdf library. Real-world PDFs are messy: some are scanned, some are damaged, and some have unusual internal structure. The extraction is therefore written to be tolerant. A file that cannot be read is recorded with the status “pdf_error” and skipped, and the batch simply continues with the next file instead of stopping. Progress is saved every 25 files, so if the machine is interrupted the job can resume from its last checkpoint rather than starting from the beginning. All three study years completed extraction with this design.

4.3 Feature Engineering

The six financial features are return on assets, the liabilities-to-assets ratio, revenue growth, operating cash flow divided by total assets, a receivables-turnover measure, and total accruals divided by total assets. These are common indicators of profitability, leverage, growth, cash-flow quality, and earnings

quality. The three text features are computed from the MD&A text: a Chinese readability measure, a sentiment balance measure, and an uncertainty ratio. The sentiment and uncertainty measures are based on a small word list prepared for the Chinese financial context, which we describe as a bootstrap dictionary because of its limited size.

Before modelling, two pre-processing rules are applied so that a few extreme numbers do not dominate the results. First, values are winsorised at the 1% and 99% levels, which means the most extreme values are pulled back to those boundaries. Second, each feature is standardised within its manufacturing sub-industry and year group, so a firm is compared with similar firms in the same year. When a group has fewer than 20 observations, we fall back to standardising over the whole sample. The same eligible observations are used to build three matrices: a finance-only matrix, a text-only matrix, and a fused matrix with all nine features. Using the same observations keeps the comparison fair.

4.4 Model Development

The primary model is Isolation Forest. The idea is simple: unusual observations are easier to separate from the rest of the data, so an observation that can be isolated with few splits receives a higher anomaly score. We set the number of trees to 500, the number of samples used to build each tree to 256, and the random seed to 42. The decision rule is fixed in advance: an observation is flagged when its score falls in the top 10% of the sample. We also check the top 5% and top 15% so that the main conclusion does not depend on one particular threshold.

As a robustness model we train an autoencoder, which is a network that compresses the features into a small middle layer and then tries to rebuild the original features. The reconstruction error, that is the difference between the rebuilt and the original values, is used as an alternative anomaly score: an observation that the model cannot rebuild well is more unusual. The autoencoder uses hidden layers of 16, 8 and 16 units, the ReLU activation function, and the Adam optimiser, and it is trained with the same fixed seed as the main model.

To explain the rankings we use SHAP, a widely used method that assigns each feature a contribution to the final score. We report three things: the overall importance of each feature, the combined importance of the financial group versus the text group, and a local explanation for the highest-ranked observations. This step is important because the output is meant to support a human reviewer, and a reviewer needs to know why a particular firm-year was ranked as unusual. The explanation only describes the model's reasoning; it does not by itself prove that any firm has misbehaved.

4.5 Evaluation Implementation

The evaluation is kept strictly separate from model training so that the benchmark cannot leak into the model. We compute the Beneish M-Score, a well-known accounting-based indicator of possible earnings manipulation, from the financial statements, and use it only as an external benchmark. We then measure how closely the anomaly scores agree with the M-Score using Spearman rank correlation. We

also compute the event rate and the enrichment ratio for the top 10% queue: the event rate is how often a high-M-Score case appears in the queue, and the enrichment ratio compares that rate with the rate in the whole sample.

For the third research question, we compare the top-ranked lists produced by the finance-only, text-only, fused and autoencoder models, and by different thresholds, and measure how much they overlap. A high overlap means the ranking is stable; a low overlap means the result is sensitive to small design choices. This comparison is computed from the same feature table, so differences come from the models rather than from different data.

4.6 Current Implementation Status and Limitations

At the time of writing, the sample construction, the download audit, the MD&A extraction for all three years, the quality-control report, and the descriptive-statistics script are complete and verified by automated tests (more than seventy tests, all passing). The one remaining input dependency is the income statement and the cash-flow statement from CSMAR, which have not yet been placed in the data folder. Once these two files are available, the financial feature table and the full experiment can be run, and the results can be written into the experiment section. The implementation described in this section will not need to change.

The main limitations are the following. First, the Chinese text dictionary is a small bootstrap version, so the sentiment and uncertainty measures are coarser than a full academic dictionary. Second, PDF extraction introduces some noise, because scanned pages and unusual layouts cannot always be read perfectly. Third, the sample is limited to manufacturing firms over three years, so the results should not be generalised to other industries or periods. Because of these limits, the pipeline should be read as a tool for ordering firms for human review, not as a proof that any specific firm has committed fraud or low-quality disclosure.

5. Experiment and Results

5.1 Final Sample and Data Coverage

The sample frame contains 3,694 manufacturing firms and 10,172 firm-years (2021: 3,165; 2022: 3,404; 2023: 3,603). After merging the six financial features, the three textual features and the Beneish M-Score, 8,140 firm-years (80.0%) across 3,265 firms have complete data and enter the main analysis (2021: 2,430; 2022: 2,769; 2023: 2,941). MD&A text availability is high: 9,991 rows are of good quality, 7 are flagged as suspicious (scanned reports or fewer than 300 tokens) and 174 have no usable text; the yearly usable shares are 96.5% (2021), 99.0% (2022) and 99.0% (2023). The download audit verified 10,051 PDFs correct out of 10,172 (98.8%); the 121 missing files are mostly pre-listing OTC (NEEQ) years of Beijing Stock Exchange firms that CNINFO does not archive.

5.2 Descriptive Statistics and Main Results (RQ1)

Table 5.1 (descriptive statistics, in the accompanying data appendix) reports the raw distributions of the nine features; the financial features are right-skewed with heavy tails, which is why the pipeline winsorises each feature at the 1st and 99th percentiles before standardisation. The benchmark (Beneish M-Score above -1.78) marks 5,110 of the 8,140 usable firm-years (62.8%) as red flags, a much higher share than in Beneish's original US sample, so it is treated only as a weak external reference, never as a training label. Table 5.2 reports the Spearman rank correlation between each ranking and the M-Score: finance-only IF 0.136 ($p < 0.001$), text-only IF -0.030 ($p = 0.008$), fused IF 0.103 ($p < 0.001$) and fused autoencoder 0.012 ($p = 0.272$). Table 5.3 reports the top-k event rates and enrichment ratios: at the top-10% threshold the finance-only enrichment is 1.217 (1.370 at top-5%) and the fused enrichment is 1.176 (1.331 at top-5%), while the text-only enrichment is 0.914 and the autoencoder 0.996. The finance-only model therefore shows a weak but highly significant positive rank agreement with the benchmark, and adding the three textual features does not improve the ranking (fused rho 0.103 < finance-only 0.136). RQ1 is answered in the negative: in this sample, textual tone features do not add incremental value over the financial features for flagging Beneish-style red flags.

5.3 Group Contributions (RQ2)

On the fused model, the financial group accounts for 65.9% of the total absolute SHAP contribution and the textual group for 34.1%. The three most important global features are ocf_to_assets (0.324), revenue_growth (0.317) and readability (0.315), so the two groups both contribute but the financial side dominates, consistent with RQ1. The top-ranked firm-year is firm 000820 (2021, sub-industry C42), with a fused anomaly score of 0.767 and a very high M-Score of 1,077.5. Its largest local SHAP driver is receivables_turnover, followed by revenue_growth and ocf_to_assets, indicating an unusually large receivables/revenue combination relative to its subindustry-year peers.

5.4 Robustness (RQ3)

The top-10% overlap between the finance-only and fused rankings is 0.828 (they largely agree because the fused model is dominated by financial features), while the finance-only versus text-only overlap is only 0.120 (the two signals rank very different firms). The autoencoder reconstruction-error ranking does not replicate the main result: its correlation with the benchmark is 0.012 ($p = 0.272$) and its top-10% enrichment is 0.996 (no better than chance). The finding is therefore sensitive to the modelling choice and should be read as a screening signal, not a stable risk label.

5.5 Discussion and Limitations

The score is a review-priority ranking signal, not a verdict on fraud. The benchmark itself is imperfect in this setting: the -1.78 cut-off flags 62.8% of the sample, so the enrichment ratios are diluted relative to a low-incidence benchmark. Known approximations include bootstrap-sized sentiment and uncertainty dictionaries, substring counting, exclusion of scanned or unreadable reports, sparse

subindustry-year standardisation fallback, and the use of the M-Score as a proxy for disclosure risk. These are disclosed in the reproducibility guide.

6. Expected Outputs, Limitations, and Ethical Boundary

The expected outputs are a reproducible company-year feature table, three comparable anomaly rankings, benchmark-association results, robustness summaries, global feature-importance outputs, and local review reports. Several limitations should be stated up front: the sample is confined to manufacturing and to 2021-2023, the external benchmarks are incomplete, text extraction and language measurement may introduce error, and unsupervised rankings are sensitive to sample composition and configuration. Throughout the project we keep data versions, extraction dates, sample rules, feature definitions, software settings, random seeds, and result tables. Anything released at company level is labelled a research-oriented review signal rather than a statement of wrongdoing.

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